

The steady three-dimensional compressible Navier–Stokes equations for an ideal gas are expressed as:

$$\nabla \cdot (\rho \mathbf{u}) = 0, \quad (1)$$

$$\nabla \cdot (\rho \mathbf{u} \mathbf{u} + p \mathbf{I} - \boldsymbol{\tau}) = 0, \quad (2)$$

$$\nabla \cdot [\mathbf{u}(E + p) - \boldsymbol{\tau} \cdot \mathbf{u} - \mathbf{q}] = 0. \quad (3)$$

where the primitive variables are the density $\rho(x, y, z)$, velocity $\mathbf{u} = (u, v, w)$, and pressure $p(x, y, z)$. The viscous stress tensor and heat flux vector are defined as

$$\boldsymbol{\tau} = \mu \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right), \quad (4)$$

$$\mathbf{q} = -k \nabla T, \quad k = \frac{\mu c_p}{\text{Pr}}, \quad c_p = \frac{\gamma}{\gamma - 1} R. \quad (5)$$

The viscosity μ may be either constant ($\mu = 1/\text{Re}$) or follow the nondimensional Sutherland law:

$$\mu(T) = \mu_0 \frac{T^{3/2}(1 + S)}{T + S}.$$

The neural network predicts $\{\rho, u, v, w, p\}$ at collocation and boundary points. The total loss combines the PDE residuals and boundary constraints.

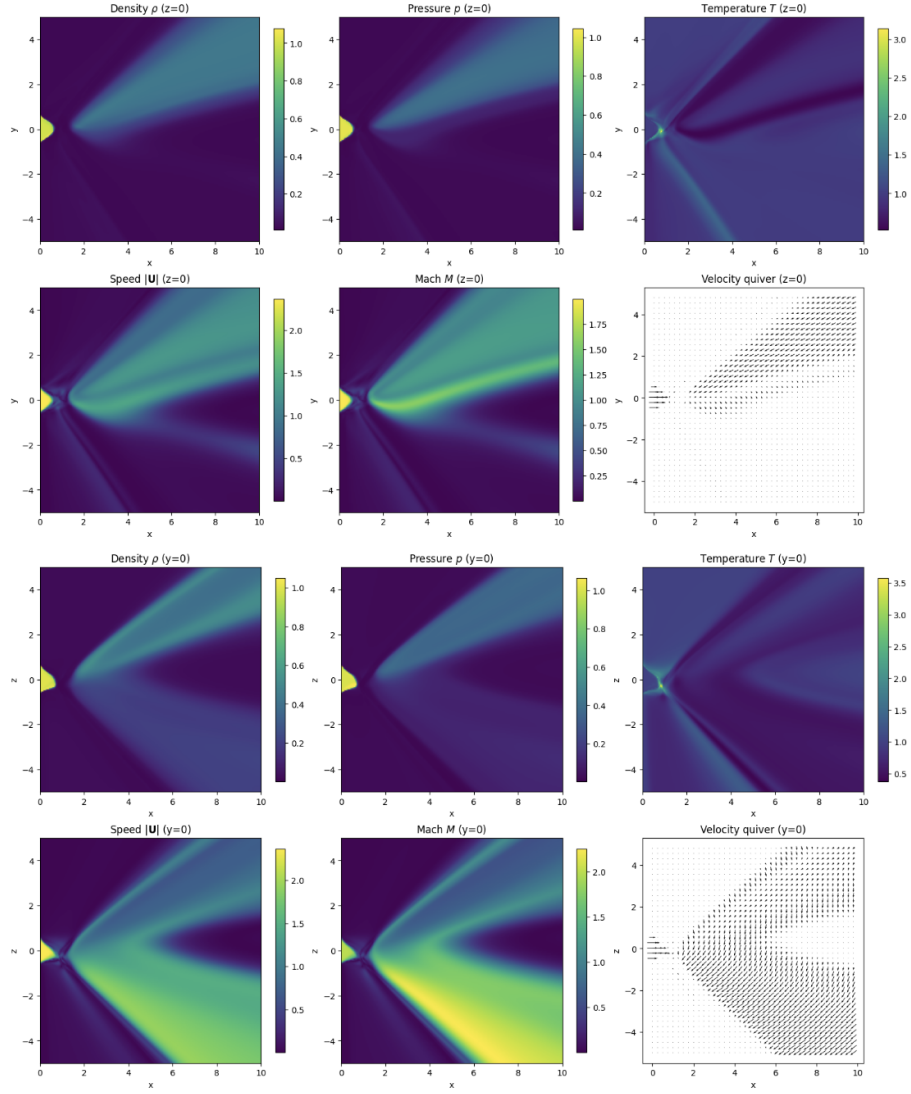


Figure 1: